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ABSTRACT

Mineral dissolution exists in numerous subsurface environments and energy engineering applications. Unlike the dissolution of pure substances, in realistic scenarios, this process is influenced by the type and distribution of minerals, thereby changing the local dissolution rate. On the contrary, local dissolution disparity gives rise to flow and solute transport heterogeneity. Such heterogeneity significantly influences processes like diagenesis, unconventional oil and gas reservoir development, and carbon storage. However, the quantification of mineral dissolution disparity in relation to flow and transport heterogeneity is relatively understudied. Numerical simulators are powerful tools for investigating reactive transport. Common approaches include continuum-scale simulations and pore-scale simulations. However, continuum-scale simulations often overlook the heterogeneity within the pore scale, whereas pore-scale simulations struggle with heavy computational costs. In this study, we applied the micro-continuum method with an extension that incorporates multiple reactive minerals, i.e., calcite and dolomite. After validating the simulator, we studied mineral dissolution in fractured media under different transport conditions and with varying mineral portions. We observed that the mineral distribution significantly changes the flow and solute transport patterns and also influences the media's characteristics, such as the permeability–porosity relationship. Furthermore, we defined a heterogeneity index to quantify the non-uniformity of permeability caused by dissolution. Our findings indicate that despite the homogeneous distribution of minerals, variations in reaction rates produce a heterogeneous permeability field following dissolution. This study demonstrates a quantitative framework for studying the synergy between multiple minerals' dissolution and reactive transport in porous media.

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I. INTRODUCTION

Mineral dissolution exists in numerous subsurface environments, such as diagenesis,[1] underground aquifers,[2] sedimentary basins,[3,4] and geothermal reservoirs.[5] Among which, one area attracts special attention: as conventional oil and gas resources become increasingly depleted, unconventional resources such as shale, tight carbonates, and tight sandstones, previously deemed non-viable for development, have emerged as key focus areas in the energy sector over the past few decades.[6–8] Unconventional reservoirs are characterized by low porosity and permeability, resulting in high resistance to fluid flow. Consequently, acid fracturing techniques are commonly employed to create artificial fractures, thereby reducing flow resistance near the wellbore. During acidization, acid solutions are introduced into low-permeability rock formations or zones near the wellbore that have been damaged during production or injection while maintaining pressures below the formation fracture threshold.[9] The injected acid then reacts with the surrounding carbonate minerals, such as calcite and dolomite, leading to the competitive dissolution of reactive minerals and the formation of high-permeability channels.[10–12] The dissolution significantly reduces the flow resistance of the injected or produced fluids, thereby enhancing the recovery efficiency of unconventional reservoirs.[13,14]

Therefore, studying the reaction and transport process in porous media of rock reservoirs is of great significance to oil and gas production engineering. In realistic scenarios, this process is affected by the type and distribution of minerals, which in turn alters the local dissolution rate. Different types of minerals possess distinct chemical and physical properties, and their distribution within a geological

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formation can vary widely. Some minerals may be more reactive under certain conditions, while others might be more resistant. This variance in mineral types can lead to a complex interaction that modifies the local dissolution rate.[15–18] On the contrary, the difference in local dissolution leads to heterogeneity in flow and solute transport. When dissolution rates differ from one location to another, it creates an uneven distribution of dissolved substances. This disparity can cause variations in the porosity and permeability of the rock matrix, which are crucial factors determining fluid flow. Areas with higher dissolution rates may have more interconnected pores, allowing fluids to flow more easily, while those with lower dissolution rates may have restricted flow paths.

Numerical solvers have emerged as effective tools in the exploration of reactive transport. Two prevalent numerical approaches are continuum-scale simulations and pore-scale simulations. Continuum-scale simulations treat the medium as a continuous substance, where properties like porosity, permeability, and reaction rates are averaged or upscaled over a representative volume.[19,20] However, continuum-scale models are less effective in accounting the intricate heterogeneity that exists at the pore scale such as mineral distribution. However, pore-scale simulations have been developed with the aim of capturing the intricate details of the pore structure. They can accurately model various processes, including the flow of fluids around individual mineral grains and the chemical reactions that take place at the fluid–solid interfaces.[21–24] However, in order to accurately represent the complex geometry of the pores, a high-resolution computational grid is necessary, which in turn requires a large number of computational grids/elements. This makes pore-scale simulations resource-intensive, potentially limiting their widespread application due to the computational demands they impose.

The micro-continuum method (MCM), is derived by volume-averaging Navier–Stokes equations over a control volume containing both fluid and solid.[20,25,26] These equations consist of a Stokes-like momentum equation weighted by porosity, with an additional drag term that accounts for the interaction between fluid and solid within the control volume. The reaction is then modeled by variation of the local solid fraction. From a geochemical perspective, a significant advantage of the micro-continuum reactive transport models is their ability to integrate high-resolution geochemical and mineralogical data with spatially resolved physical properties that control fluid flow.[27] Further research is needed to understand the intricate relationships between various minerals and their combined effects on reactive transport. For instance, different minerals may possess distinct chemical and physical properties, which can lead to diverse reaction kinetics characteristics, thereby influencing the overall flow patterns and solute transport mechanisms.[28] Thus, in this study, we will focus on the application of the MCM framework, studying reactive transport in system consisting of multiple minerals, with special attention on fractured media.

Fractures are ubiquitous in geological formations, such as carbonates and shales, which also exhibit complex mineralogical compositions. Fractures are inherently anisotropic and heterogeneous features that significantly influence flow and solute transport within the medium. Numerous studies have demonstrated the substantial impacts of fractures[22,29,30] on reactive transport. Consequently, we incorporate fractures into our study and aim to determine how fractured media induce heterogeneity in the reactive system.

Previous studies have focused extensively on macroscopic fracture morphology evolution and transport characteristics. For instance, an acidizing fracturing model based on a dual-scale continuum approach can characterize the effects of mineral compositions on acid-etched fracture morphology in carbonate rocks.[31] Furthermore, pore-scale Darcy-Brinkman-Stokes (DBS) reactive transport modeling reveals that fracture configurations (length/density/connectivity) and mineral compositions (calcite/clay/quartz) significantly influence porosity–permeability relationships and reactive transport efficiency.[32] Nevertheless, there is still a lack of a macroscopic scale on the cross-scale correlations between pore structure evolution and seepage response. This study systematically uncovers multiscale permeability evolution mechanisms in porous media by developing multiscale coupling algorithms and integrating stochastic geometric modeling of fracture networks under *in situ* geological conditions.

Mineral grains were generated using the quartet structure generation set (QSGS). The QSGS method is a random generation-growth algorithm designed for porous media generation.[33] It utilizes four parameters to control the micro-scale structures of the media, thereby enabling the creation of porous media with tailored characteristics. To generate media with fractures, we adopt two types of algorithms: (1) For idealized fracture generation, we defined a start node, an end node, and the segment that connects them. This approach provides a simplified yet effective way to model fractures within the porous media. (2) For the mapped fractures, we sourced the fracture geometry from digital rock images. Subsequently, we used tensor operations to map this geometry onto the simulation domain. This approach enables us to incorporate realistic fracture patterns observed in actual rock samples into our simulations, enhancing the authenticity and accuracy of the modeled media.

Quantitative characterization of reservoir heterogeneity is critical for hydrocarbon development. The Dykstra–Parsons coefficient (D–P coefficient), proposed in 1950, quantifies heterogeneity based on a log-normal permeability distribution assumption by calculating the relative difference between the median permeability (K_{50}) and the 84.1st percentile permeability ($K_{84.1}$).[34] However, it neglects spatial configurations and the influence of low-permeability zones. Breakthrough coefficients (maximum/average permeability) and permeability range (maximum/minimum permeability), developed in 2009, provide clear engineering significance.[35] While existing methods improve the systematic assessment of permeability distribution, they still fail to adequately capture local heterogeneity. Current metrics largely rely on oversimplified single-distribution assumptions, in contrast to the complex reality of reservoirs, which often feature multiscale heterogeneity, spatially dependent attributes, and distinct fracture–matrix interactions. These limitations underscore the critical need to develop innovative permeability evaluation metrics that account for such multidimensional complexity.

Permeability heterogeneity is a crucial aspect of reservoir rock properties, reflecting not only the variability in permeability values but also the spatial distribution's irregularity. To better capture this characteristic, traditional single-value metrics may fall short. Kitanidis introduced an innovative approach that utilizes the second spatial moment to quantify solute concentration dispersion.[36] This method combines both concentration values and spatial locations through the calculation of the second spatial moment, providing a more comprehensive description of heterogeneity. We find that this quantification method

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can also be applied to characterize permeability heterogeneity. By integrating both the magnitude of physical properties and their spatial coordinates, this method offers a holistic measure of both numerical and spatial heterogeneity. This approach not only allows us to quantify the variability in permeability values but also accurately captures the spatial distribution of permeability. Consequently, it provides a more refined and comprehensive tool for analyzing permeability distributions at the pore scale in reservoir rocks. Compared to traditional single-value metrics, this combined consideration of magnitude and spatial location offers significant advantages in characterizing permeability heterogeneity.

This manuscript is structured as follows: Sec. I concisely outlines the motivation, methods, and techniques employed in this study. Section II elaborates on the formulation and methodologies used to investigate reactive transport, mineral formation, and fracture development. In Sec. III, we present validation cases, single- and multi-fracture simulations that illustrate the influence on fluid flow and solute transport. Additionally, we conduct a quantitative analysis to determine how the mineral fraction affects the permeability–porosity relationship. Also, we investigate mineral content's influence on the heterogeneity of the permeability in the vicinity of fractures. This research presents a quantitative framework for the study of the synergy existing between the dissolution of multiple minerals and reactive transport within porous media.

II. METHODS

A. Micro-continuum method

We adopt the micro-continuum method (MCM) developed and validated in previous studies.[25,26,37–40] The MCM uses volume fraction to indicate the solid/fluid phases,

$$\varepsilon_i = \frac{V_i}{V_{tot}}, \quad i = \{f, s\}, \tag{1}$$

where $V_f(\mathrm{m}^3)$, $V_s(\mathrm{m}^3)$, and $V_{tot}(\mathrm{m}^3)$ are the volume of the fluid phase, solid phase, and the total volume. $\varepsilon_f(-)$ and $\varepsilon_s(-)$ are the volume fraction of the fluid and solid phases, respectively. The flow velocity is governed by

$$\frac{1}{\varepsilon_f}\left[\frac{\partial(\rho_f \mathbf{u})}{\partial t} + \nabla\cdot\left(\frac{\rho_f}{\varepsilon_f}\mathbf{u}\mathbf{u}\right)\right] = -\nabla P + \frac{\mu_f}{\varepsilon_f}\nabla^2\mathbf{u} - \frac{\mu_f}{k}\mathbf{u}, \tag{2}$$

$$\nabla\cdot\mathbf{u} = R\left(\frac{1}{\rho_f} - \frac{1}{\rho_s}\right), \tag{3}$$

where $\mathbf{u}(\mathrm{m\,s^{-1}})$ is the fluid velocity, $\rho_f(\mathrm{kg\,m^{-3}})$ and $\rho_s(\mathrm{kg\,m^{-3}})$ are the fluid and solid density, $P(\mathrm{kg\,m^{-1}\,s^{-2}})$ is the pressure, $\mu_f(\mathrm{kg\,m^{-1}\,s^{-1}})$ is the dynamic viscosity, $R(\mathrm{kg\,m^{-3}\,s^{-1}})$ quantifies the consumption due to local reaction, and $k(\mathrm{m}^2)$ is the local permeability that is defined by the Kozeny–Carman equation,

$$k = k_0\frac{\varepsilon_f^3}{(1-\varepsilon_f)^2}, \tag{4}$$

where $\sqrt{k_0}$ is defined as the characteristic length of the micro-structure.[25] Solute concentration (e.g., H^+) is solved by the micro-continuum advection-diffusion-reaction equation (ADRE),

$$\frac{\partial(\varepsilon_f C)}{\partial t} + \nabla\cdot(\mathbf{u}C) = \nabla\cdot\left(\varepsilon_f D_e\nabla C\right) - \frac{\xi}{M_e}R, \tag{5}$$

where $C(\mathrm{mol\,m^{-3}})$ is the solute concentration, $D_e(\mathrm{m^2\,s^{-1}})$ is the effective dispersivity, $\xi(-)$ is the stoichiometric coefficient, and $M_e(\mathrm{kg\,mol^{-1}})$ is the effective molar mass of the solid phase. The reaction, either dissolution or precipitation, is quantified by

$$\rho_s\frac{\partial\varepsilon_s}{\partial t} = -R \tag{6}$$

and

$$\frac{\partial\varepsilon_s}{\partial t} = \alpha 10^{b\cdot T}A_s(\varepsilon_{s,eq} - \varepsilon_s)\gamma_{H^+}C, \tag{7}$$

$$R = \alpha 10^{b\cdot T}A_s(\varepsilon_s - \varepsilon_{s,eq})\gamma_{H^+}C, \tag{8}$$

where $\alpha(\mathrm{s^{-1}})$ is the reactive constant, $T(\mathrm{K})$ is the temperature, $b = 0.022\,\mathrm{K^{-1}}$ is a constant coefficient quantifying how the reaction rate is influenced by geological formation temperature[41] and $\gamma_{H^+} = 1\times10^{-3}\,\mathrm{m^3\,mol^{-1}}$ is the activity. $\varepsilon_{s,eq}$ is the equilibrium solid fraction and $A_s = \nabla\varepsilon_f/|\nabla\varepsilon_f|(-)$ is the local surface norm. To model dissolution, the equilibrium solid fraction is set as $\varepsilon_{s,eq} = 0$. We consider dissolution is caused by acid and the local pH is computed by

$$pH = -\log_{10}(\gamma_{H^+}C). \tag{9}$$

B. Multiple minerals and fracture generation

The two-dimensional porous domain with multiple mineral grains is generated using the quartet structure generation set (QSGS) method. Four phases are generated: (1) calcite, (2) dolomite, (3) inert matrix, and (4) pore space including pores and initial fractures. The media generation is started with random number generation, that creates grain core points. Their locations within the grid are determined in accordance with the preset volume fraction and core probability. These points are then scattered across the grid. To guarantee a proper distribution of grains, further sieving is carried out to ensure that the core points adhere to the stipulated grain spacing requirements.

The growth of grains initiates from the core points and extends into the adjacent grid areas, following the predefined growth and directional probabilities (Fig. 1). During this expansion, the influence of grain shape is factored in. Shape files are provided to fine-tune the form and orientation of the grains. Moreover, the interaction between neighboring grains is accounted for; when such interactions occur, the growth probability is diminished under the control of specific parameters. This iterative progression persists until the quantity of generated grains aligns with the targeted solid volume fraction.

Once the grains have been generated, they are categorized into two mineral phases, namely mineral I (calcite) and mineral II (dolomite), based on the pre-established mineral phase ratios. A random sampling approach is adopted to allocate a certain number of grains to mineral I, while the leftover grains are assigned to mineral II.

As shown in Fig. 1, a schematic diagram of heterogeneous minerals is presented at the center, with the realistic model generated by the QSGS method showcased on the left and a dissolution schematic simulated using the DBS method illustrated on the right.

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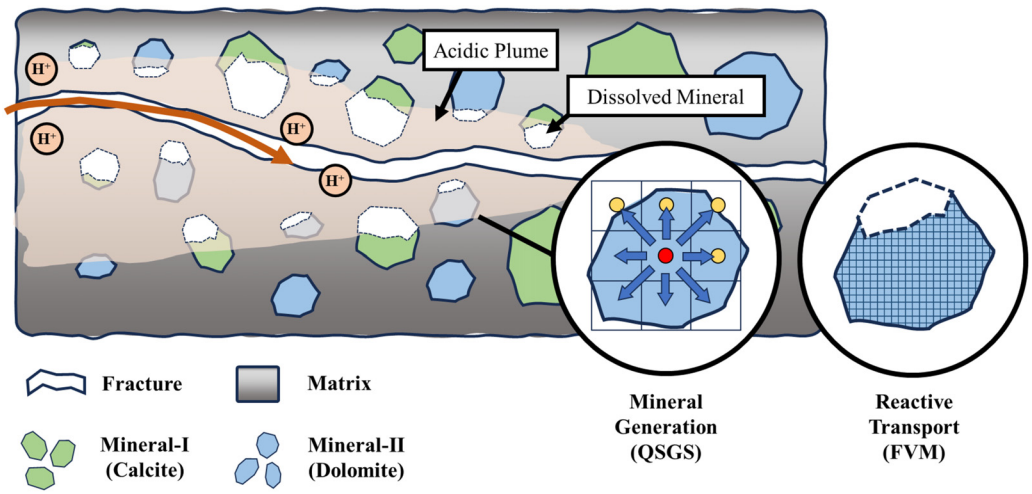


FIG. 1. Schematic diagram of a medium containing multiple minerals (e.g., calcite and dolomite) and fractures undergoing acid dissolution. The spatial distribution of the minerals is generated using the quartet structure generation set (QSGS) method, while the reactive transport process is simulated using the micro-continuum method (MCM), implemented via the finite volume method (FVM).

C. Fracture generation

To generate a comprehensive and accurate representation of fractures and fracture networks, which are crucial for understanding various geological and engineering processes such as fluid flow in media, we adopted two distinct types of fractures: (1) idealized fracture and (2) mapped fracture (Fig. 2).

The idealized fractures are defined by a series of precisely located nodes. These nodes are arranged in a particular order, forming a framework upon which the fracture is constructed. Once the nodes are determined, a line segment is used to connect these nodes, thus creating the physical manifestation of the fracture. Moreover, an essential parameter, the fracture aperture, is also accurately defined. The fracture aperture represents the width of the fracture, which significantly influences fluid flow characteristics within the fracture.

The mapped fractures, on the contrary, rely on digital rock images. Digital rock images offer a high resolution and detailed view of the internal structure of rocks, where fractures can be identified. We extract fracture topology from the digital rock images. During this extraction process, we define the inlet and outlet of each fracture. Subsequently, a series of geometric transformation operations are applied. Translation is used to shift the position of the fracture within

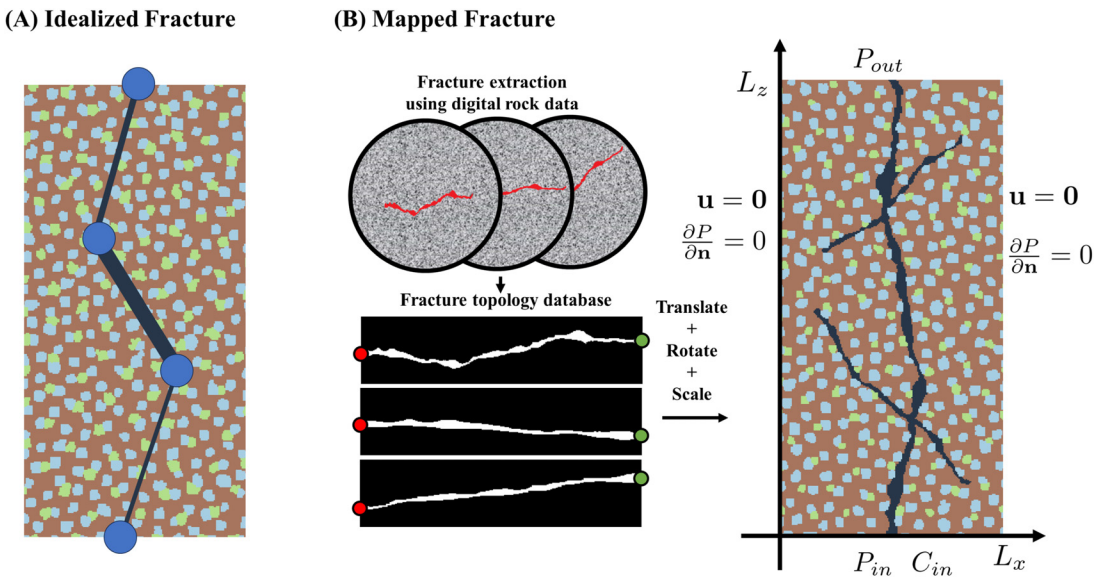


FIG. 2. Fracture generation algorithm: (a) Idealized fracture, defined by the nodes and segments connecting nodes and (b) mapped fracture, using fracture topology imported from digital rock images, and mapped to domain using tensor operation.

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the coordinate system. Rotating the fracture allows us to change its orientation, mimicking the natural variation in fracture angles found in real-world rocks. The scaling operation, however, adjusts the size of the fracture, enabling us to represent fractures of different lengths and widths. By applying these translation, rotating, and scaling operations, we map the fracture topology, which includes information about the fracture's shape, its connection with other fractures (if any), and the overall spatial arrangement, to the numerical domain. The numerical domain is generated using the QSGS. During the mapping process, we have the flexibility to adjust the length and orientations of the fractures.

D. Reactive transport simulation

In the numerical simulations, the governing equations, Eqs. (3)–(7), require the specification of appropriate initial and boundary conditions for velocity, pressure, and concentration fields, as shown in Figs. 3 and 5. The flow is imposed by either a constant pressure drop condition or a constant inlet velocity condition, $P(z=0)=P_{in}$ and $P(z=L_z)=P_{out}$ are the pressures at the inlet and outlet, respectively. The simulation domain is two-dimensional with physical dimensions of L_x and L_z. The bottom boundary is defined as the inlet, the top boundary as the outlet, and the lateral boundaries are set to no-flow conditions. For the concentration field, a fixed concentration $c=C_0$ is applied at the inlet, while all other boundaries are subject to zero-gradient conditions.

The simulation parameters are as follows: the length of a single grid cell is set to 1×10^{-6}m, the computation domain contains 100 000 to 500 000 cells, and the time step is 0.1 s. The matrix's initial permeability (K_{matrix}) is $1\times10^{-16}\text{m}^2$. The fluid has a dynamic viscosity of 1×10^{-6}Pas and a density of $1000\,\text{kg/m}^3$, with the diffusion coefficient being $1\times10^{-13}\,\text{m}^2/\text{s}$. The solid phase includes two reactive minerals: calcite with a density of $2710\,\text{kg/m}^3$ and dolomite with a density of $2860\,\text{kg/m}^3$. Their relative molecular masses are 100 and 184 g/mol, respectively.

The numerical solver, pre-processing and post-processing modules, and interfaces for different physical processes were implemented using a combination of the open-source computational fluid dynamics platform OpenFOAM, Bash scripting, C++, and Python, running on a Linux-based system. Parallel computing was utilized to enhance computational efficiency.

This study adopts a micro-continuum approach, utilizing the open-source solvers porousMedia4Foam[41] and hybridPorousInterFoam.[26] Building on the original solver's capability for single reactive solids, we developed a novel multi-mineral reactive module to simulate systems with heterogeneous mineral assemblages. This methodological advancement significantly expands the framework's utility for modeling geologically complex systems, such as multi-mineral reactive transport scenarios. The overall process and numerical algorithm are illustrated in Fig. 4.

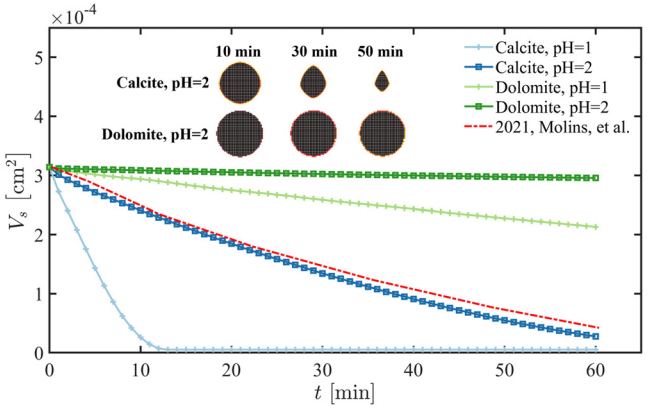


FIG. 3. Mineral dissolution simulation validation case: The blue curves represent the volume of calcite grains as a function of time, while the green curves depict the change in the volume of dolomite grains and the red dashed line is the validation case presented in the literature.[39] The shapes of the grains are presented at the top of the figure.

III. RESULT

A. Validation case

To verify the implementation and compare the dissolution rates of different mineral grains, we conduct a validation as depicted in Fig. 3. The same case, which was also utilized in a previous study,[39] is employed here. In that previous work, different reactive simulators were validated by comparing them against this identical case. We consider two types of minerals: calcite and dolomite. The dissolution rate of calcite is greater than that of dolomite. For the values of these dissolution rates, we adopt the same calcite dissolution rate used in the work by Molins *et al.*,[39] which are presented in Table I.

Four cases are examined in this study: the dissolution of calcite grains when the inlet pH is set at 1 and 2, and the dissolution of dolomite grains under acidic conditions with the same pH values of 1 and 2. The outcomes of these tests are presented in Fig. 3. In the figure, the curve depicts the variation in the volume of the solid grain over time. Additionally, the shapes of the grains in the pH = 2 case are shown at three specific time points: 10, 30, and 50 min.

Upon analyzing the results, we observe that the volume of the solid decreases in all cases. A notable difference in dissolution rates is observed between calcite and dolomite. Specifically, the dissolution rate of calcite grains is significantly higher than that of dolomite grains. The high dissolution rate of calcite, combined with its interactions with the fluid flow and solute transport processes, leads to the formation of a droplet-like structure in the calcite grains. In contrast, the dolomite grains undergo a more homogeneous dissolution process, lacking the distinct morphological changes seen in the calcite grains.

B. Flow and transport influenced by dissolution

Figures 5 and 6 present the simulation results for single-fracture and multiple-fracture cases, respectively. In these cases, calcite and dolomite have an equal volumetric proportion. The fractures used in these simulations are mapped from digital rock images. Figure 5 showcases the simulation results for a single fracture embedded in a domain measuring $100\times200\,\mu\text{m}^2$. Figure 5(a) depicts the volume of the solid as a function of time under different acidic conditions and pressure drop inputs. The solid curve represents the case with a pH of 1, where the pressure drop ranges from 1 Pa to 1×10^4 Pa. The dashed line represents the case with a pH of 3, with the same range of pressure drop variation.

From the figure, it can be observed that when the pressure drop is small ($\leq 1\times10^2$ Pa), the variations in solid dissolution are similar. However, when the flow is stronger, the dissolution process is

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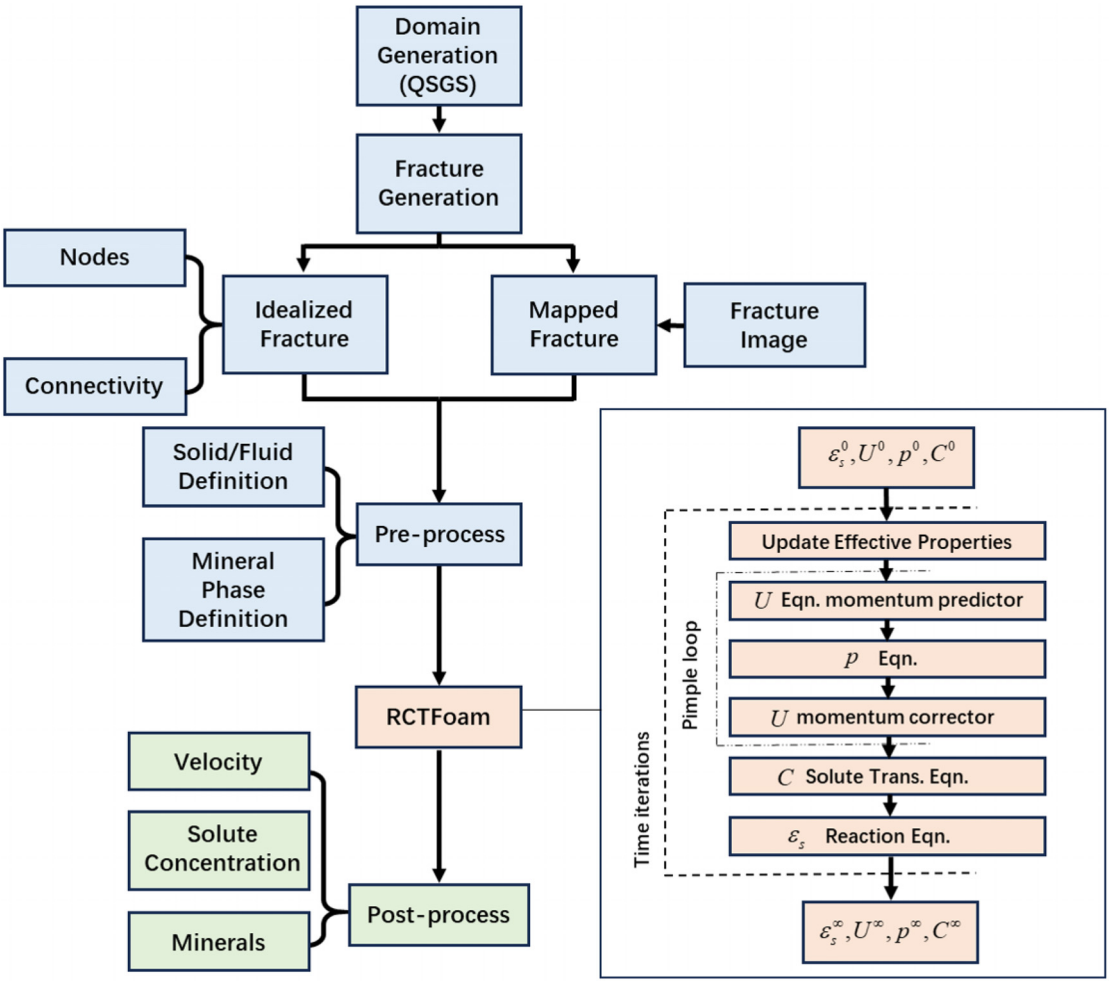


FIG. 4. Schematic diagram of the workflow and algorithm.

significantly influenced by the flow and solute transport. To conduct a more in-depth investigation, we present a non-reactive case in Fig. 5(b) and cases with a pH of 1 under different pressure drops in Figs. 5(c)–5(e). Each plot shows the pore structure after 120 min of dissolution, the local pH, and the velocity streamlines. When the flow is weak, the acidic solution remains close to the fracture, and dissolution mainly occurs near the fracture wall. As a result, the dissolution heterogeneity induced by the minerals is not very noticeable. When the flow is strengthened, the pH distribution [Fig. 5(d)] is distorted by the pores formed by the rapidly dissolved minerals. This indicates that when the flow is strong, the diversity in mineral dissolution can strongly affect the pore-scale physical processes.

TABLE I. Parameters set for the validation cases.

Case	$\alpha(\mathrm{s}^{-1})$	$C_0(\mathrm{mol\,m}^{-3})$	$U_0(\mathrm{m\,s}^{-1})$
Calcite, pH = 1	0.89×10^{-5}	1.0×10^{-4}	1.2×10^{-3}
Dolomite, pH = 1	1×10^{-6}	1.0×10^{-4}	1.2×10^{-3}
Calcite, pH = 2	0.89×10^{-5}	1.0×10^{-5}	1.2×10^{-3}
Dolomite, pH = 2	1×10^{-6}	1.0×10^{-5}	1.2×10^{-3}

To explore this process in a more complex geometry, we mapped five fractures onto a $400 \times 400\ \mu\mathrm{m}^2$ domain containing uniformly distributed calcite and dolomite grains. In Fig. 6, each row corresponds to a different pressure drop case, and each column represents the pore structure at $t = 30$ min and $t = 60$ min. When the flow is relatively weak, dissolution occurs mainly near the fractures. As the pressure increases, the solute migrates into the matrix, causing the grains with a higher dissolution rate to dissolve first. The dissolution creates pore spaces, which enhance the local permeability. This, in turn, leads to an improvement in flow and transport. Such a positive feedback mechanism results in the heterogeneity induced by the dissolution process.

C. Permeability–porosity behavior in multiple minerals case

To investigate the effect of mineral proportion and quantify permeability variation, we use an idealized single-fracture case with

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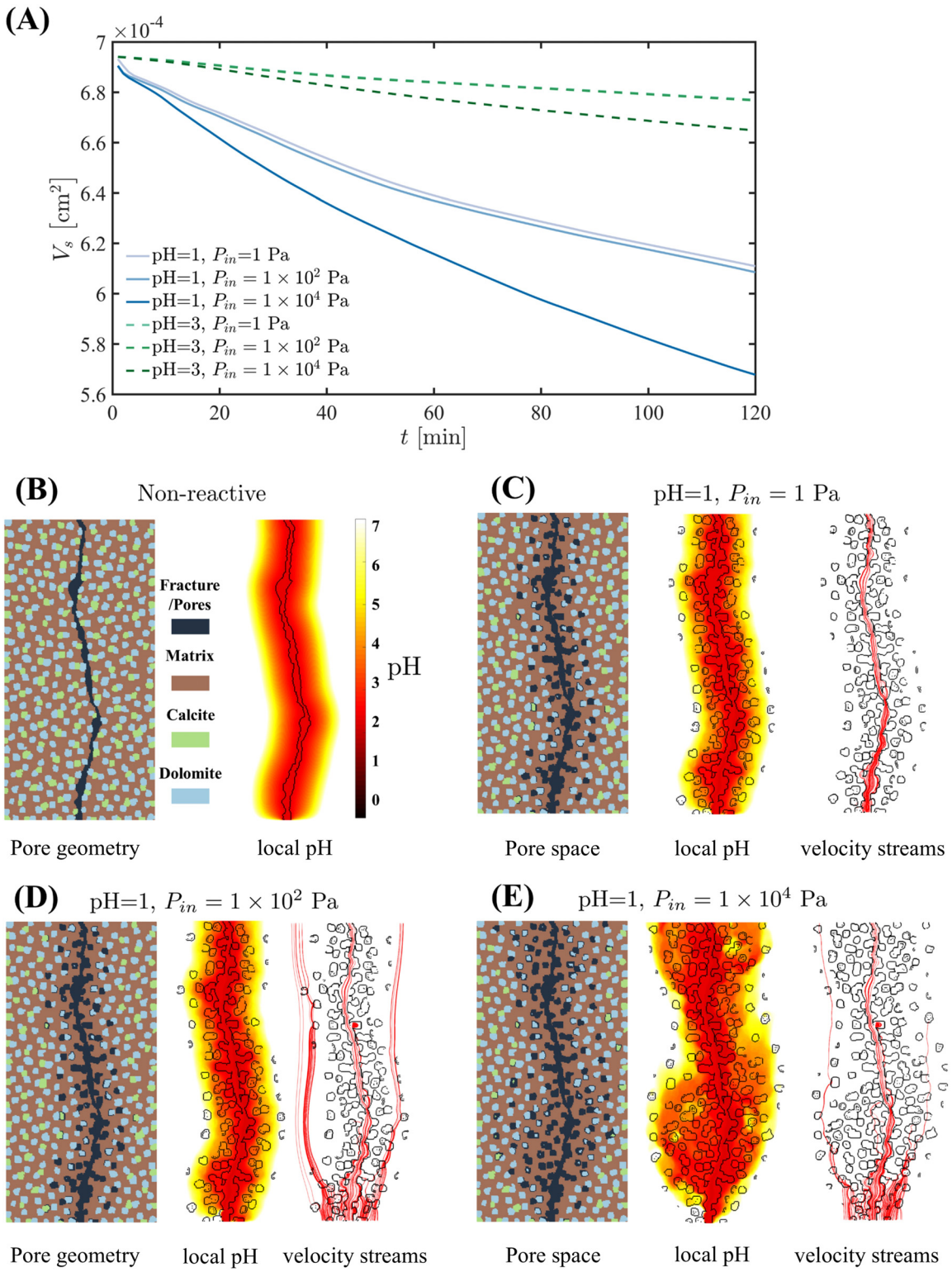


FIG. 5. (a) Simulation results of different inlet pressures (P_{in}) and pH values. (b) Non-reactive reference case. (c)–(e) Pore geometry, local pH, and velocity streams in cases where the inlet pressure ranges from 1 Pa to 1×10^4 Pa.

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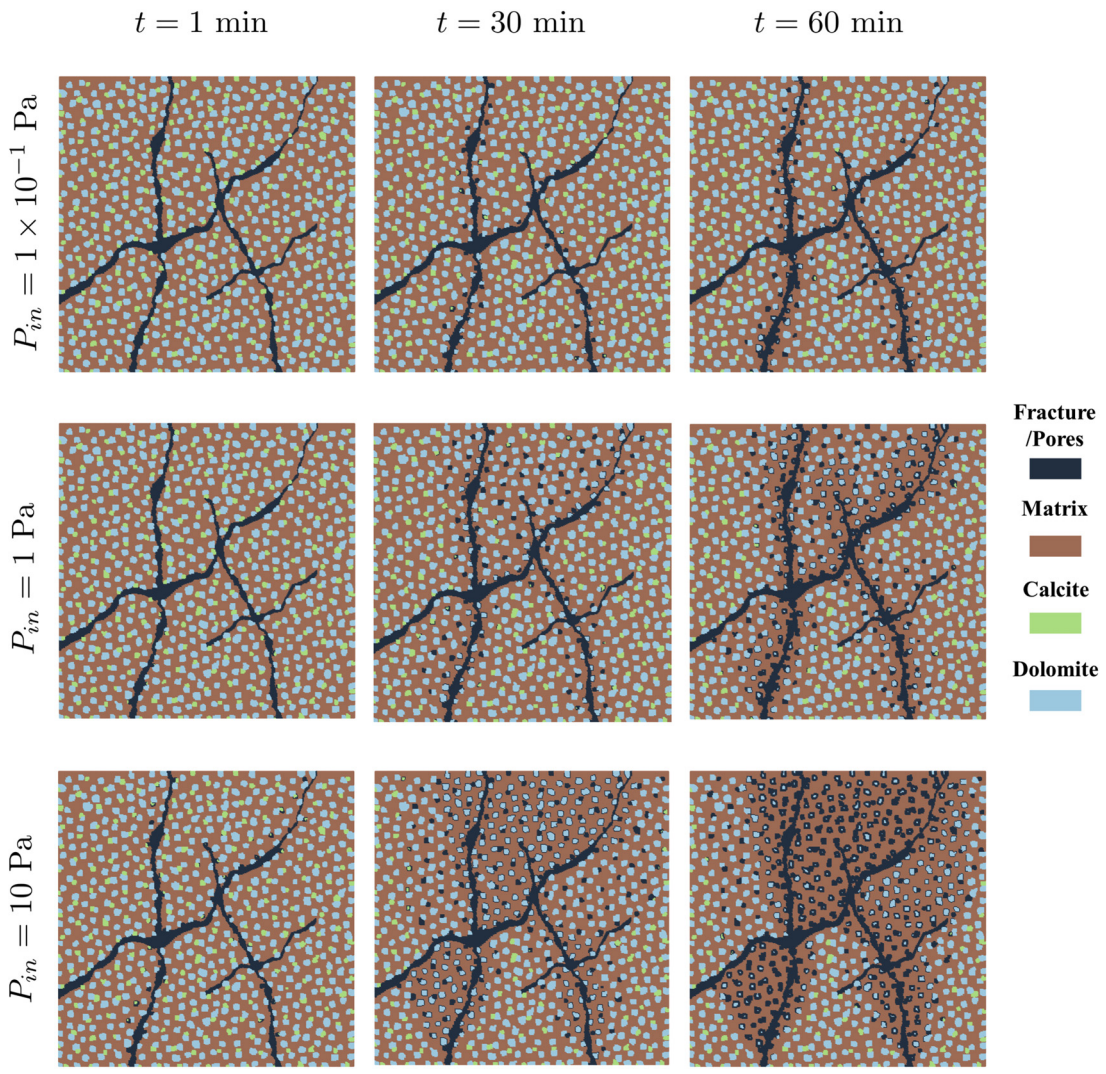


FIG. 6. Simulation results of the multiple fractures case. Each row presents the pore geometry at distinct time instances, while different rows correspond to various inlet pressure values.

different mineral grain ratios for the simulation (Fig. 7). To quantify the permeability–porosity under different simulation conditions, we compute the porosity and permeability from the simulation results,

$$\begin{cases} \langle \cdot \rangle = \dfrac{1}{V} \displaystyle\int_V \cdot dV, \\ \langle \cdot \rangle^f = \dfrac{1}{V_f} \displaystyle\int_{V_f} \cdot dV, \end{cases} \tag{10}$$

where V_f denotes the volume of fluid within the control volume V. The porosity ϕ of the porous medium is calculated from the average local porosity field,

$$\phi = \langle \varepsilon_f \rangle. \tag{11}$$

By calculating the average velocity and pressure gradient within the volume V and utilizing Darcy's law, the permeability K_{REV} at the REV scale can be derived as

$$\langle \bar{v}_f \rangle = -\frac{K_{REV}}{\mu_f} \nabla \langle \bar{p}_f \rangle^f. \tag{12}$$

Through these calculations, we obtained the relationship between permeability K_{REV} at the representative elementary volume (REV) scale and porosity under varying mineral proportions, inlet pressures, and fracture network conditions (Fig. 7). This analysis resulted in graphical representations illustrating how these parameters influence the REV-scale permeability within the computational domain.

As shown in Fig. 7, in each sub-figure, the blue and green curves correspond to pure mineral cases. The red curve is the dissolution dynamic of a calcite–dolomite mixture. We observe that, in the pure

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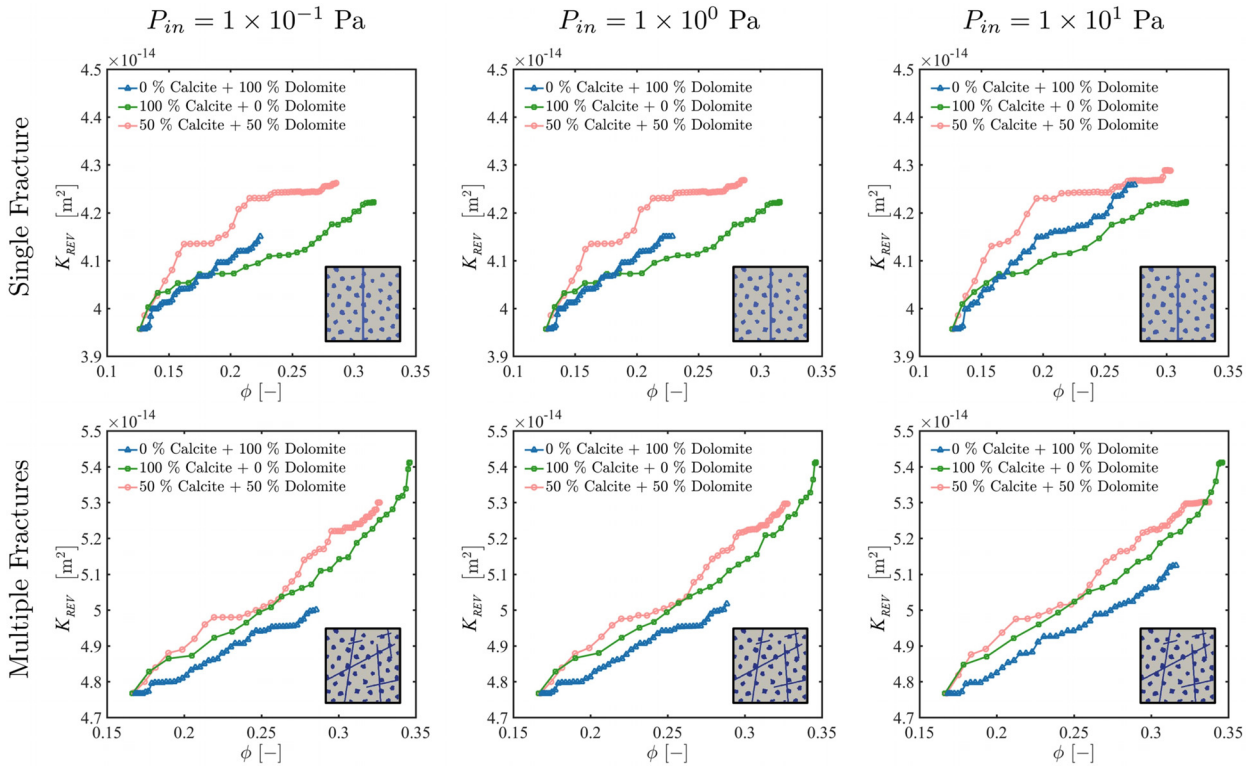


FIG. 7. Permeability at the REV scale as a function of porosity under conditions of different mineral proportions, different pressures, and varying fracture numbers.

mineral dissolution case, the permeability–porosity correlations ($K_{REV} - \phi$) are similar. In the mixed case, the $K_{REV} - \phi$ curve shows a fast increment near the low-porosity range and slower increasing rate when porosity is high. This deviation suggests that in the pure mineral case, dissolution starts near the fracture and progresses into a deeper matrix, and the permeability increment follows a similar trend. When multiple minerals exist in the system, dissolution front penetrates into the matrix faster and all the highly reactive minerals are dissolved, thus the overall permeability exhibits a fast increment. The second row of Fig. 7 shows the results of multiple ideal fractures case, we can see that the $K_{REV} - \phi$ curve shows similar trend between pure and mixed mineral cases.

Furthermore, regardless of whether the system contains a single fracture or multiple fractures, within the simulated pressure range and at the stages of reactive transport development, systems with two minerals exhibited higher permeability than systems with a single mineral for the same porosity. This finding indicates that the interaction between different mineral types plays a crucial role in influencing the overall permeability of the medium.

This phenomenon may be intrinsically linked to the distribution of permeability fields at the pore scale. Specifically, the presence of multiple reactive minerals likely leads to a more complex and interconnected pore structure, facilitating improved fluid flow and mass transport. This enhanced permeability can be attributed to the development of preferential flow paths and variations in pore connectivity.

D. Permeability heterogeneity

To further investigate local permeability change and distribution of preferential flow zones, we divide the computational domain into sub-domains based on the following steps: (1) A linear grid was defined, and each grid cell was traversed. For each grid cell, the x, z, and permeability K values of all data points within the cell were accumulated and divided by the number of data points in the cell, thereby calculating the average x-coordinate, z-coordinate, and permeability value for each grid cell. (2) Based on the computed average x-coordinate x_{Sub}, average z-coordinate z_{Sub}, and corresponding average permeability value K_{Sub}, a color map of the grid was generated, where the color intensity represents the magnitude of the permeability K. This method allows for a pixelate representation of the permeability distribution at the pore scale and aids in analyzing the underlying mechanisms of permeability variation in systems with different mineral compositions and fracture distributions (Fig. 8).

Mineral heterogeneity influences the reaction rate by accelerating local reactions or altering the reaction sequence. From the pore-scale permeability distribution maps, it can be observed that when two minerals with different reaction rates are present, the distribution of high-permeability zones becomes more dispersed. This is particularly evident along the main flow direction, where the "spread" of permeability is more extensive.

The heterogeneity of minerals results in a non-uniform reaction process. In a scenario where the mineral composition is assumed to be entirely homogeneous, such as entirely composed of dolomite or

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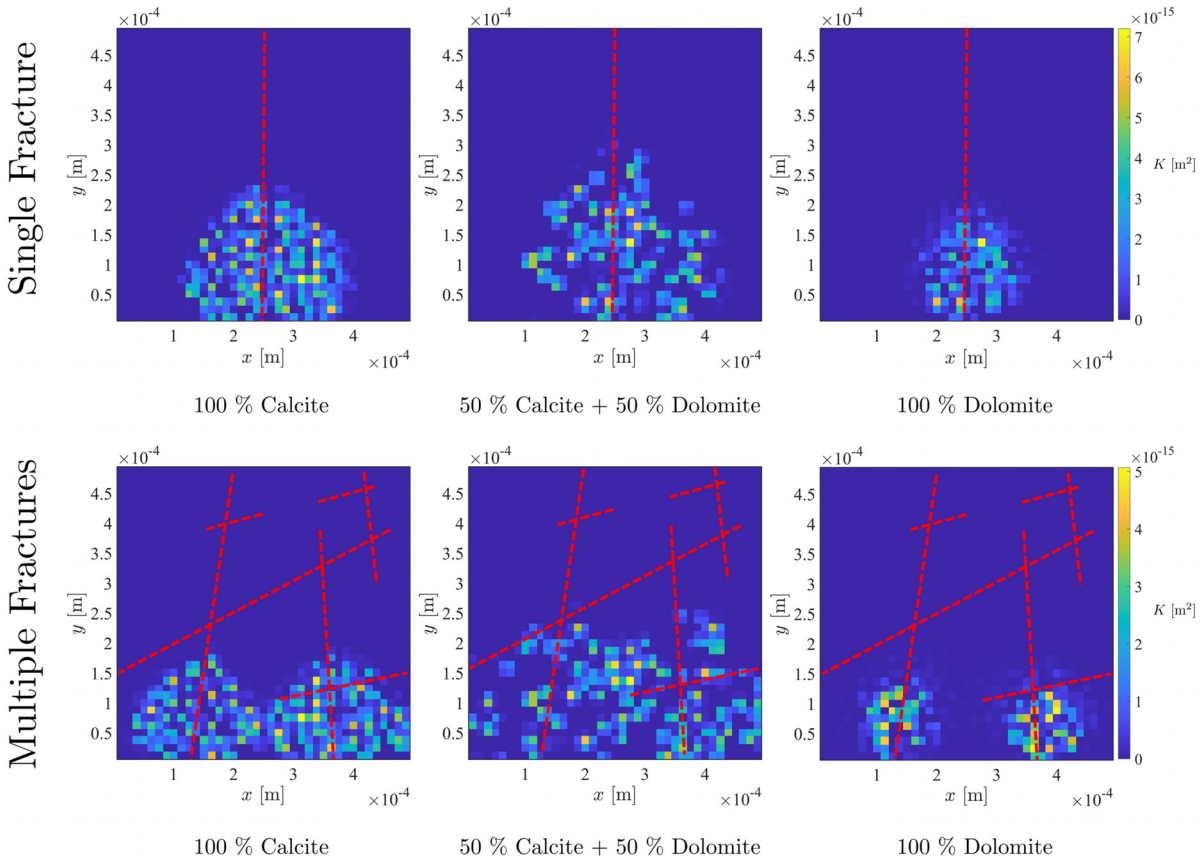


FIG. 8. The distribution of pore-scale permeability under conditions of varying crack numbers and different mineral proportions.

calcite, the reaction would proceed gradually from the bottom to the top. The reaction front would progressively advance, and mineral grains would gradually dissolve into the fluid phase. Reaction products would accumulate in specific regions and migrate as the reaction continues.

In contrast, in the case of heterogeneous mineral composition with components that have varying reaction rates, the process behaves differently. The rapid dissolution of faster-reacting minerals (e.g., calcite) preferentially enhances local permeability, forming high-conductivity flow channels. These channels dominate fluid transport and isolate regions containing slower-reacting minerals (e.g., dolomite) due to bypassing effects, significantly retarding their dissolution. For example, once calcite-rich zones develop into dominant flow paths, dolomite adjacent to the flow channels or in the surrounding matrix may experience delayed reaction initiation due to limited reactant transport to these isolated regions.

This non-uniform reaction progression highlights the impact of mineral heterogeneity on the overall behavior of reactive transport in porous media. By calculating the second spatial moment of permeability under different mineral composition conditions, it is observed that the permeability distribution becomes more dispersed after the reaction in porous media with minerals exhibiting heterogeneous reaction rates. This indicates an increase in permeability heterogeneity.

IV. HETEROGENEITY INDEX

To assess the heterogeneity induced by dissolution, we quantify the permeability field using the spatial moments analysis. This method was applied to analyze how concentration distributes and the dispersive effect on the heterogeneity of the solute plume.[36]

First, we define normalized permeability at each grid cell using the total permeability across the domain,

$$p(x,z,t) = \frac{k(x,z,t)}{\int_\Omega k(x,z,t)dA}, \tag{13}$$

where Ω is the entire domain, and $k(x,z,t)$ represents the permeability at location (x,z) at time t. $p(x,z,t)$ has the unit (m^{-2}), and it can be interpreted as the probability distribution of permeability at a given position and time. The first spatial moment is used to represent the centroid of the field, while the second spatial moment describes the mean squared distance from the center in both the x- and z-directions. When second spatial moment is small the field is closer to a circular shape, and the distribution is more homogeneous, when the second spatial moment is larger, the field exhibits a more spreading and dispersive distribution. The first spatial moment is

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$$\begin{cases} \alpha_{\perp}(t) = \int_A p(x,t)x dA, \\ \alpha_{\parallel}(t) = \int_A p(z,t)z dA. \end{cases} \tag{14}$$

The subscript $\perp$ indicates it is the direction perpendicular to the main flow (i.e., x-direction) and $\parallel$ denotes the direction parallel to the main flow (i.e., z-direction). The second spatial moment is used to quantify the heterogeneity of the permeability distribution,

$$\begin{cases} \Gamma_{\perp}(t) = \int_A p(x,z,t)(x-\alpha_{\perp})^2 dA, \\ \Gamma_{\parallel}(t) = \int_A p(x,z,t)(z-\alpha_{\parallel})^2 dA. \end{cases} \tag{15}$$

A larger $\Gamma_{\perp}(t)$ indicates greater heterogeneity in the x-direction, and similarly, a larger $\Gamma_{\parallel}(t)$ signifies greater heterogeneity in the z-direction. Figure 9 is the result of Γ for different calcite content and different Péclet number, which is defined as

$$Pe = \frac{U_{in} b}{D_e}. \tag{16}$$

The darker curve corresponds to a higher Pe number. We can see that Γ shows a peak when calcite has an intermediate percentage. The reason behind this phenomenon can be attributed to the interaction between minerals with different reaction kinetics. When fast-reacting and slow-reacting minerals coexist in the porous medium, the fast-reacting minerals dissolve more rapidly, creating preferential flow channels. These channels accelerate fluid flow and enhance mass transport, which, in turn, further isolates the regions containing slower-reacting minerals. This isolation not only delays the dissolution of slow-reacting minerals but also results in uneven pore space development. Over time, this selective dissolution process leads to a highly heterogeneous permeability distribution, as preferential pathways dominate the flow field. Moreover, the behavior differs between the directions perpendicular and parallel to the main flow.

In the direction perpendicular to the main fracture flow, the heterogeneity index of the permeability distribution decreases as the Pe number increases. This suggests that higher flow rates lead to reduced heterogeneity, likely because faster flow enhances mass transport efficiency, leading to a more uniform distribution of dissolution products and reduced spatial variability. In contrast, in the direction parallel to the main flow, there is less variation in permeability, and heterogeneity is observed with changes in the Pe number. This result underscores that the heterogeneity in mineral reaction rates is the dominant factor influencing the heterogeneity of the post-reaction permeability distribution. This phenomenon can be attributed to the structural changes induced by differential mineral dissolution rates. Fast-reacting minerals tend to create preferential flow paths early in the process, which subsequently isolates regions where slow-reacting minerals are located. This not only increases the complexity of the permeability distribution but also enhances the dispersivity of the pore network, leading to further heterogeneity in the porous medium.

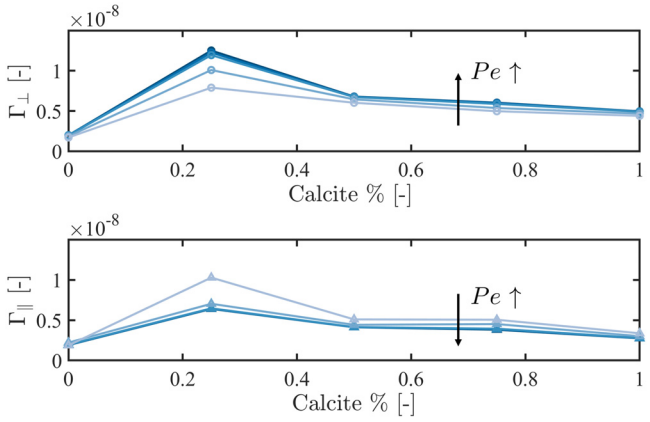


FIG. 9. The variation of the heterogeneity index with changes in mineral proportions and Pe. The darker curve represents a higher Pe number case.

V. CONCLUSION

To quantitatively assess the heterogeneity of mineral distribution and its impact on pore media during the reaction process, this study enhances the existing micro-continuum method (MCM). Modifications were applied to the original open-source code by incorporating heterogeneous minerals and variable reaction coefficients, enabling comprehensive numerical simulations. Furthermore, the original quartet structure generation set (QSGS) algorithm was adapted to generate digital rock models that more accurately reflect the complexity of real subsurface heterogeneous environments. A novel heterogeneity index was introduced to improve the quantification and comparison of mineral heterogeneity and the resultant permeability heterogeneity following reactions.

The findings of this analysis reveal several critical insights: First, the presence of fractures within the porous medium introduces significant heterogeneity in the pore structure, which persists even in scenarios where mineral distribution is uniform. This inherent heterogeneity leads to notable variations in the effective permeability of the medium, particularly at the scale of the representative elementary volume (REV). In contrast to systems composed of a single mineral type, an uneven mineral composition results in increased permeability at the REV scale. This phenomenon is intricately linked to the distribution of permeability at the pore scale; as this distribution becomes increasingly dispersed, the overall level of heterogeneity intensifies, thereby enhancing permeability at the REV scale.

The simulation results indicate that the observed increase in permeability is primarily driven by the formation of preferential flow paths through rapid dissolution of highly reactive minerals (e.g., calcite). However, these pathways dominate fluid transport and isolate regions containing slower-reacting minerals (e.g., dolomite) via bypassing effects, significantly retarding their dissolution. This selective isolation amplifies the permeability contrast between high-conductivity channels and low-permeability regions, ultimately enhancing the overall heterogeneity of the porous matrix. Consequently, the coupled effects of flow focusing and mineral dissolution not only localize solute transport within high-conductivity channels but also induce spatial changes in the pore structure. This interplay further complicates the permeability landscape of the medium, highlighting the reactive transport coupling dynamics governing heterogeneous porous systems.

In conclusion, the incorporation of mineral heterogeneity and the resulting formation of preferential flow paths are crucial factors that govern the reactivity and transport processes within porous media. Future research should further explore these interactions and their implications for subsurface flow and reactive transport modeling.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xingming Du: Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **Enhao Liu:** Data curation (equal); Formal analysis (equal); Investigation (equal); Visualization (equal). **Runqing Shan:** Conceptualization (equal); Investigation (equal). **Fengchang Yang:** Conceptualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Bowen Ling:** Conceptualization (equal); Project administration (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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